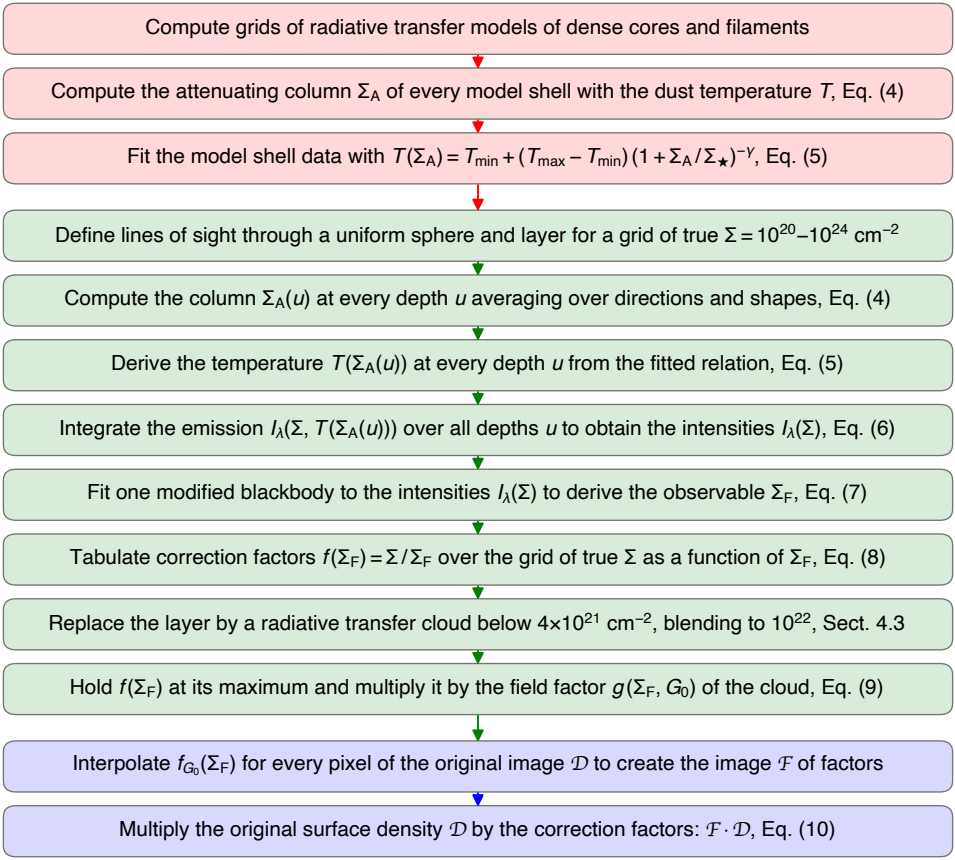
Compute grids of radiative transfer models of dense cores and filaments
Compute the attenuating column Σ_A of every model shell with the dust temperature T, Eq. (4)
Fit the model shell data with $T(\Sigma_A) = T_{min} + (T_{max} - T_{min})\,(1 + \Sigma_A / \Sigma_\star)^{-\gamma}$, Eq. (5)
Define lines of sight through a uniform sphere and layer for a grid of true $\Sigma = 10^{20}$–10^{24} cm^{-2}
Compute the column $\Sigma_A(u)$ at every depth u averaging over directions and shapes, Eq. (4)
Derive the temperature $T(\Sigma_A(u))$ at every depth u from the fitted relation, Eq. (5)
Integrate the emission $I_\lambda(\Sigma, T(\Sigma_A(u)))$ over all depths u to obtain the intensities $I_\lambda(\Sigma)$, Eq. (6)
Fit one modified blackbody to the intensities $I_\lambda(\Sigma)$ to derive the observable Σ_F, Eq. (7)
Tabulate correction factors $f(\Sigma_F) = \Sigma / \Sigma_F$ over the grid of true Σ as a function of Σ_F, Eq. (8)
Replace the layer by a radiative transfer cloud below $4{\times}10^{21}$ cm^{-2}, blending to 10^{22}, Sect. 4.3
Hold $f(\Sigma_F)$ at its maximum and multiply it by the field factor $g(\Sigma_F, G_0)$ of the cloud, Eq. (9)
Interpolate $f_{G_0}(\Sigma_F)$ for every pixel of the original image $\mathcal{D}$ to create the image $\mathcal{F}$ of factors
Multiply the original surface density $\mathcal{D}$ by the correction factors: $\mathcal{F} \cdot \mathcal{D}$, Eq. (10)